

PRAKTIKUM SISTEM OPERASI

MODUL 6

Sekuensial dan konkuren



Oleh:

(Fahmi Hasan Asagaf)

(2311104074)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

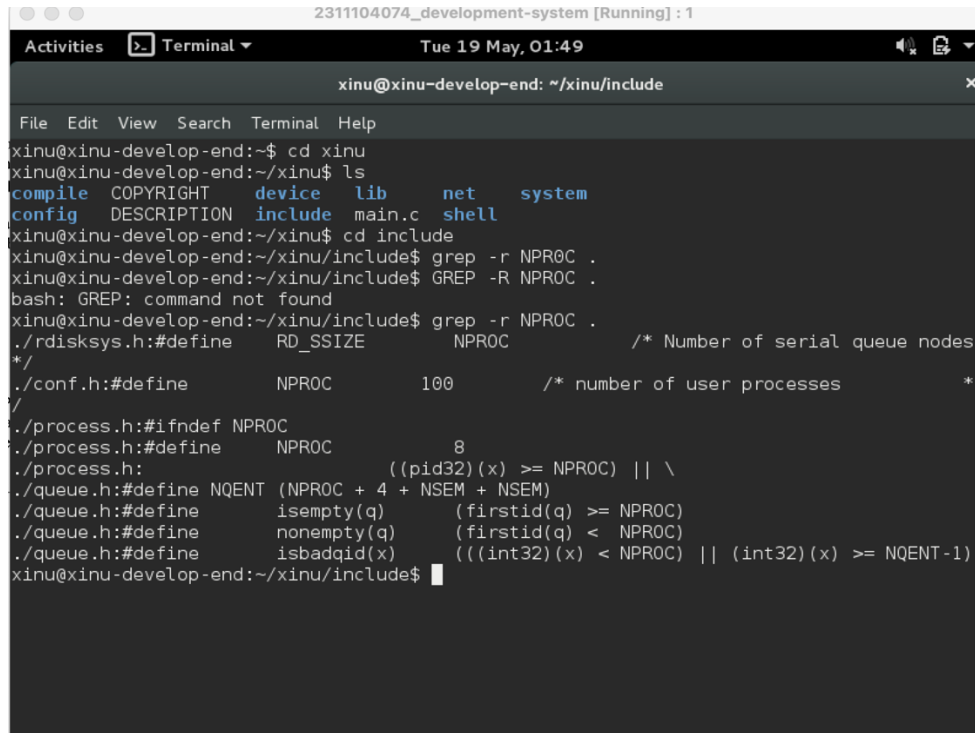
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL/TP

1.



The screenshot shows a terminal window titled "2311104074_development-system [Running] : 1". The terminal has a menu bar with "Activities", "Terminal", and a dropdown arrow. The title bar indicates the current directory is "~/xinu/include". The terminal content shows the following sequence of commands and output:

```
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ ls
compile  COPYRIGHT  device  lib      net      system
config  DESCRIPTION  include  main.c  shell
xinu@xinu-develop-end:~/xinu$ cd include
xinu@xinu-develop-end:~/xinu/include$ grep -r NPROC .
xinu@xinu-develop-end:~/xinu/include$ GREP -R NPROC .
bash: GREP: command not found
xinu@xinu-develop-end:~/xinu/include$ grep -r NPROC .
./rdisksys.h:#define      RD_SSIZE      NPROC      /* Number of serial queue nodes
*/
./conf.h:#define          NPROC          100          /* number of user processes      *
/
./process.h:#ifndef NPROC
./process.h:#define      NPROC          8
./process.h:              ((pid32)(x) >= NPROC) || \
./queue.h:#define NQENT (NPROC + 4 + NSEM + NSEM)
./queue.h:#define      isempty(q)      (firstid(q) >= NPROC)
./queue.h:#define      nonempty(q)     (firstid(q) < NPROC)
./queue.h:#define      isbadqid(x)    (((int32)(x) < NPROC) || (int32)(x) >= NQENT-1)
xinu@xinu-develop-end:~/xinu/include$
```

```
2311104074_development-system [Running] : 1
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
GNU nano 2.2.6 File: conf.h Modified

/*
 *
 */
#define Nlfs 1
#define Nlfl 6
#define Nnam 1
#define Npipem 1
#define Npip 10

#define NDEVS 35

/* Configuration and Size Constants */
#define NPROC 150 /* number of user processes */
#define NSEM 100 /* number of semaphores */
#define IRQBASE 32 /* base ivec for IRQ0 */
#define IRQ_TIMER IRQ_HW5 /* timer IRQ is wired to hardware 5 */
#define IRQ_ATH_MISC IRQ_HW4 /* Misc. IRQ is wired to hardware 4 */
#define CLKFREQ 200000000 /* 200 MHz clock */

#ifndef ETHER0
#define ETHER0 0
#endif

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
Left
```

Pada Xinu, batas maksimal proses terdapat pada file conf.h dengan deklarasi:

```
#define NPROC 100
```

Artinya maksimal proses pada sistem adalah 100 proses.

Kemudian diubah menjadi:

```
#define NPROC 150
```

Sehingga sistem dapat membuat maksimal 150 proses.



```
2311104074_development-system [Running] : 1
Activities Terminal Tue 19 May, 02:35
xinu@xinu-develop-end: ~/xinu/system
File Edit View Search Terminal Help
GNU nano 2.2.6 File: main.c

/* main.c - main */
#include <xinu.h>

void prosesA(void)
{
    kprintf("Proses A berjalan\n");
}

void prosesB(void)
{
    kprintf("Proses B berjalan\n");
}

process main(void){

    resume(create(prosesA,1024,20,"A",0));
    resume(create(prosesB,1024,20,"B",0));

    return OK;
}

[ Read 23 lines ]
^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
Left %
```

2311104074_development-system [Running] : 1

Activities Terminal Tue 19 May, 02:32

xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

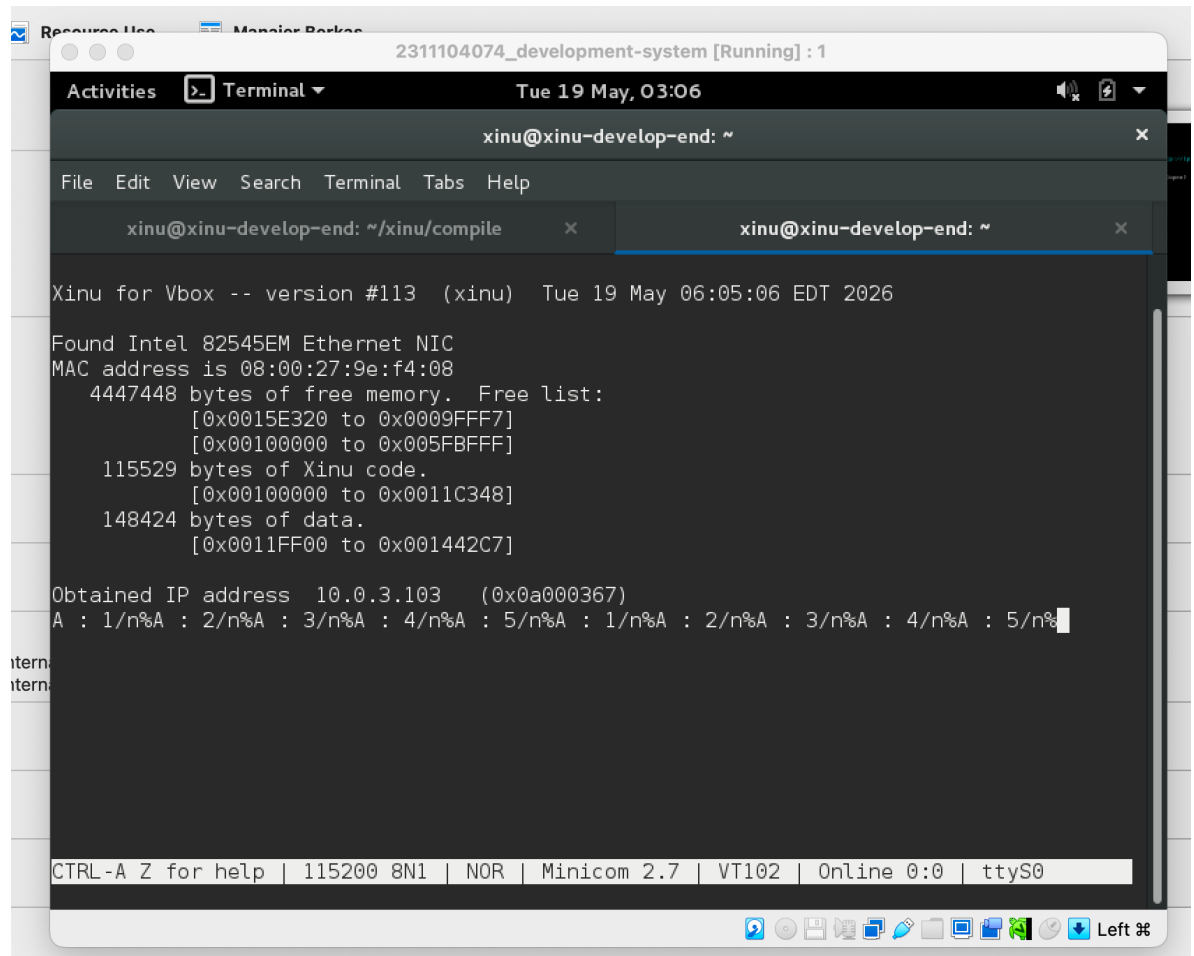
```
Xinu for Vbox -- version #110 (xinu) Tue 19 May 05:27:14 EDT 2026
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
  4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
  115497 bytes of Xinu code.
    [0x00100000 to 0x0011C328]
  148424 bytes of data.
    [0x0011FF00 to 0x001442C7]

Obtained IP address 10.0.3.103 (0x0a000367)
Proses A berjalan
Proses B berjalan
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

Left %

2. output



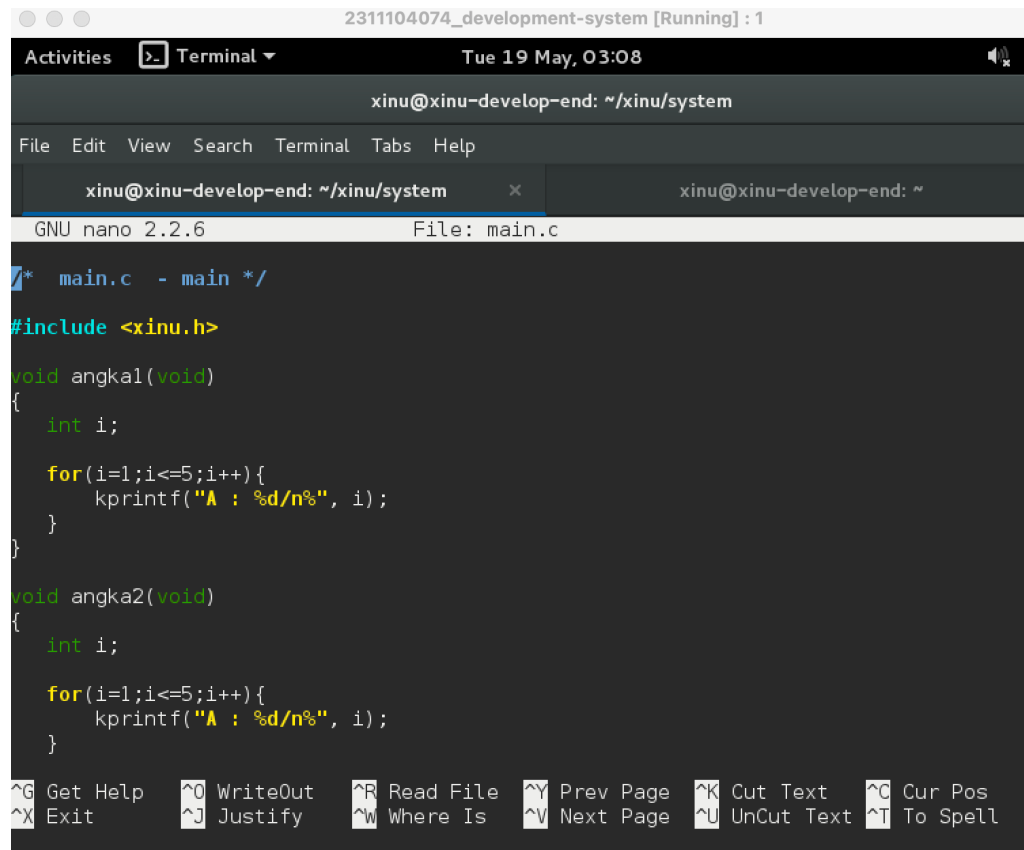
The screenshot shows a terminal window titled "2311104074_development-system [Running] : 1". The terminal displays the output of the Xinu boot process. The output includes the version number (#113), the date and time (Tue 19 May 06:05:06 EDT 2026), the discovery of an Intel 82545EM Ethernet NIC with MAC address 08:00:27:9e:f4:08, the amount of free memory (4447448 bytes), the size of the Xinu code (115529 bytes), and the size of the data (148424 bytes). It also shows the obtained IP address (10.0.3.103) and a series of prompts for user input (1/n%A : 2/n%A : 3/n%A : 4/n%A : 5/n%A : 1/n%A : 2/n%A : 3/n%A : 4/n%A : 5/n%A :). The terminal window has a menu bar with "File", "Edit", "View", "Search", "Terminal", "Tabs", and "Help". The status bar at the bottom shows "CTRL-A Z for help", "115200 8N1", "NOR", "Minicom 2.7", "VT102", "Online 0:0", and "ttyS0".

```
Xinu for Vbox -- version #113 (xinu) Tue 19 May 06:05:06 EDT 2026
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
115529 bytes of Xinu code.
    [0x00100000 to 0x0011C348]
148424 bytes of data.
    [0x0011FF00 to 0x001442C7]

Obtained IP address 10.0.3.103 (0x0a000367)
A : 1/n%A : 2/n%A : 3/n%A : 4/n%A : 5/n%A : 1/n%A : 2/n%A : 3/n%A : 4/n%A : 5/n%A :

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

3Source code

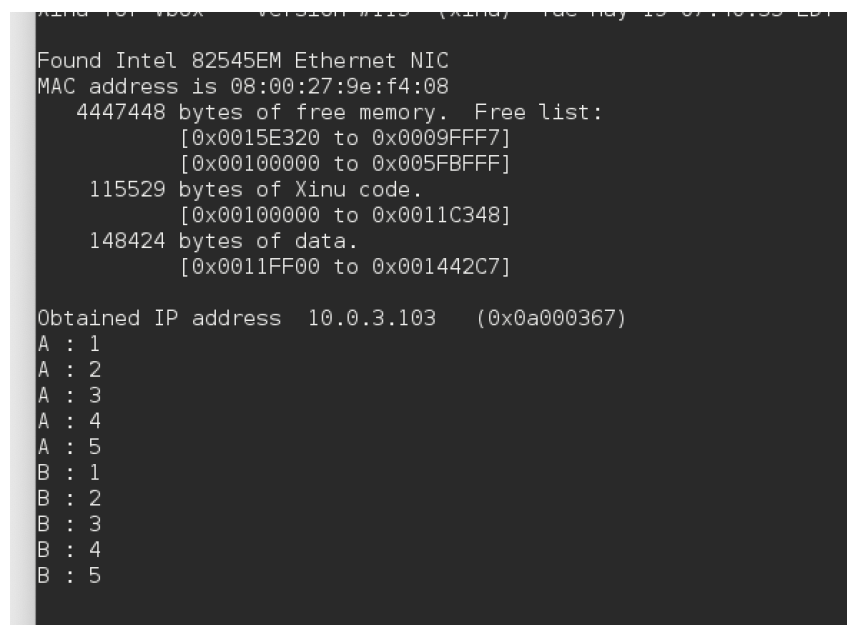


The screenshot shows a terminal window titled "2311104074_development-system [Running] : 1". The terminal is running the GNU nano 2.2.6 editor, editing the file "main.c". The code in the editor is as follows:

```
/* main.c - main */  
  
#include <xinu.h>  
  
void angka1(void)  
{  
    int i;  
  
    for(i=1;i<=5;i++){  
        kprintf("A : %d/n%", i);  
    }  
}  
  
void angka2(void)  
{  
    int i;  
  
    for(i=1;i<=5;i++){  
        kprintf("A : %d/n%", i);  
    }  
}
```

The bottom of the terminal window shows a status bar with various keyboard shortcuts: ^G Get Help, ^O WriteOut, ^R Read File, ^Y Prev Page, ^K Cut Text, ^C Cur Pos, ^X Exit, ^J Justify, ^W Where Is, ^V Next Page, ^U UnCut Text, and ^T To Spell.

output



The screenshot shows the output of the program. It starts with a message "Found Intel 82545EM Ethernet NIC" and "MAC address is 08:00:27:9e:f4:08". It then displays the amount of free memory and the free list, followed by the amount of Xinu code and data. Finally, it shows the obtained IP address and the output of the two functions, which are "A : 1", "A : 2", "A : 3", "A : 4", "A : 5", "B : 1", "B : 2", "B : 3", "B : 4", and "B : 5".

```
Found Intel 82545EM Ethernet NIC  
MAC address is 08:00:27:9e:f4:08  
4447448 bytes of free memory. Free list:  
[0x0015E320 to 0x0009FFF7]  
[0x00100000 to 0x005FBFFF]  
115529 bytes of Xinu code.  
[0x00100000 to 0x0011C348]  
148424 bytes of data.  
[0x0011FF00 to 0x001442C7]  
  
Obtained IP address 10.0.3.103 (0x0a000367)  
A : 1  
A : 2  
A : 3  
A : 4  
A : 5  
B : 1  
B : 2  
B : 3  
B : 4  
B : 5
```

4.sourcecode

```
GNU nano 2.2.6                                     File: ms
/* main.c - main */

#include <xinu.h>

/*
 * Nama : Fahmi Hasan Asagaf
 * NIM : 2311104074
 */

void proses1(void)
{
    while(1){
        kprintf("Proses 1 berjalan\n");
    }
}

void proses2(void)
{
    while(1){
        kprintf("Proses 2 berjalan\n");
    }
}

process main(void)
{
    resume(create(proses1,1024,20,"P1",0));
    resume(create(proses2,1024,20,"P2",0));

    return OK;
}
```

output

[illegible]

5.source code

```
/* main.c - main */

#include <xinu.h>

/*
Nama : Fahmi Hasan Asagaf
NIM : 2311104074
*/

int32 n = 0;

void produser(void)
{
    int32 i;

    for(i=1; i<=1000; i++){
        n++;
    }
}

void konsumer(void)
{
    int32 i;

    for(i=1; i<=1000; i++){
        kprintf("Nilai dari n adalah %d\n", n);
    }
}

process main(void)
{
    resume(create(produser, 1024, 20, "produser", 0));
    resume(create(konsumer, 1024, 20, "konsumer", 0));

    return OK;
}
```

Output

```
xinu@xinu-develop-end: ~/xinu/system
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ cd system
xinu@xinu-develop-end:~/xinu/system$ nano main.c
xinu@xinu-develop-end:~/xinu/system$ sudo minicom
[sudo] password for xinu:

Xinu for Vbox -- version #116 (xinu) Tue May 19 07:47:16 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
115561 bytes of Xinu code.
    [0x00100000 to 0x0011C368]
148424 bytes of data.
    [0x0011FF00 to 0x001442C7]

Obtained IP address 10.0.3.103 (0x0a000367)
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
```

Hasil program tidak sesuai intuisi karena proses produser dan konsumen berjalan secara konkuren dan menggunakan variabel global yang sama tanpa sinkronisasi. Saat produser menaikkan nilai n, konsumen dapat membaca nilai n pada waktu yang bersamaan sehingga terjadi race condition. Akibatnya output yang tampil tidak selalu berurutan dan dapat berubah-ubah setiap program dijalankan.